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JAVA PROJECT REPORT ON

PERSONAL HEALTH RECORD MONITORING SYSTEM

Submitted by

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DECLARATION

I hereby declare that the project entitled "**PERSONAL HEALTH RECORD MONITORING SYSTEM"** submitted in partial fulfillment of the requirements for the course **JAVA PROGRAMMING** is a **record of my original work** carried out under the guidance of **MUGILAN R**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

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Selvamalini s

CERTIFICATE

This is to certify that the project entitled **"PERSONAL HEALTH RECORD MONITORING SYSTEM"** is a **bona fide work** carried out by **SELVAMALINI I-B.SC IT-'B',INFORMATION TECHNOLOGY**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **JAVA PROGRAMMING**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place:

Date:

Submitted for the Viva-Voce Examination held on _____

Guide:
(Technical Trainer, KGCAS)

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ABSTRACT

The **Personal Health Record Monitoring System** is a Java-based application designed to automate the process of recording, managing, and monitoring personal health data for individuals and healthcare providers. In many households and clinics, health records are maintained manually, which can lead to data errors, delays in retrieving medical history, and inefficient health management. This project aims to simplify the health monitoring process by providing an automated system that records health parameters, analyzes trends, and generates an accurate health summary report.

The system allows users to input daily health readings such as blood pressure, glucose level, heart rate, and body temperature, and automatically analyzes the recorded data including alerts for abnormal or critical health values. The application is developed using **Core Java programming concepts**, including classes, objects, methods, conditional statements, loops, and exception handling. The project can be implemented either as a **console-based application or using Java Swing for a graphical user interface.**

The main objective of the system is to improve health monitoring efficiency, reduce human errors in data recording, and provide faster access to patient health information. The system generates a clear and structured health report that displays the recorded parameters, analysis results, alerts, and overall health status.

This project demonstrates the practical implementation of **object-oriented programming concepts in Java** and provides a basic foundation for developing more advanced personal healthcare management systems in the future.

1. INTRODUCTION

In the healthcare industry, efficient management of personal health records plays a crucial role in monitoring patient well-being and ensuring timely medical assistance. Traditionally, many individuals and healthcare providers rely on manual record-keeping methods, where health data is maintained using handwritten notes or physical files. This manual process is time-consuming and can often lead to data loss, errors in medical history, and difficulty in tracking health progress over time.

With the advancement of computer techonology,automated health record systems have become essential for improving efficiency and accuracy in healthcare management.A computerized monitoring system helps patients and doctors quickly access medical records,track health parameters automatically,and maintain organized records of diagnoses and treatments.

The **Personal Health Record Monitoring System** is a Java-based application designed to automate the process of health data management and report generation. The system allows the user to input health parameters such as blood pressure, glucose levels, and heart rate, and automatically analyze trends over time.

This project is developed using Core Java programming concepts such as classes, objects, methods, conditional statements, loops, and exception handling. The application can be implemented either as a console-based program or using Java Swing for a graphical user interface(GUI).

1.1Objective

The main objective of the **Personal Health Record Monitoring System** is to develop a simple and efficient health monitoring application using Core Java that automates the process of health data recording and report generation.

The specific objectives are:

*To design and develop a java-based health record monitoring system
*To automatic the tracking of health parameters and trends

*To reduce manual errors in health data management

*To provide a user-friendly system for patients and doctors

*To demonostrate Core java Programming concepts such as classes,objects,and loops

*To generate a clear and organized health report for users

1.2 Overview

The Personal Health Record Monitoring System is a menu-driven application developed using Java programming language. The system provides a digital platform where patients and healthcare providers can input health parameters, track medical history, and monitor health trends automatically.

The system displays a list of health metrics along with their recorded values. The user can enter daily health readings such as blood pressure, heart rate, glucose level, and body temperature. The program stores and analyzes each parameter and generates a complete health summary report including alerts for abnormal values

The system helps simplify the health monitoring process by eliminating manual record-keeping and reducing the chances of data errors. The application can be developed as a console-based program or a graphical user interface using Java Swing.

This project demonstrates the practical use of Java programming in developing real-world applications for personal healthcare management and remote patient monitoring.

1.2 Features

The Personal Health Record Monitoring System includes the following main feat

Patient Profile Management – Stores and displays personal details of the patient including name, age, and medical history.

Health Parameter Input– Allows the user to enter daily health readings such as blood pressure, glucose level, heart rate, and temperature.

Automatic Health Analysis – Automatically analyzes recorded values and flags abnormal or critical health readings.

Alert Generation – Generates alerts and warnings when health parameters exceed safe or normal threshold levels

Report Generation – Produces a detailed health summary showing parameter history, trends, dates, and overall health status.

User-Friendly Interface – Provides a simple and easy-to-use menu-driven interface for patients and doctors.

Fast Record Access – Reduces the time required to retrieve and review a patient's health history.

2. PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied in detail and the requirements of the system are identified. This stage helps in understanding how the system should work and what features are required to solve the problem efficiently.

The analysis phase focuses on identifying the current issues in the existing system and proposing a computerized solution that improves efficiency and accuracy.

2.1 Problem Definition

In many households and healthcare settings, personal health records are often maintained manually using paper files or simple notebooks. This traditional method creates several challenges

Some common problems include:

• Errors in recording health parameters
• Time-consuming data retrieval process
• Difficulty in tracking health trends over time
• Lack of proper alerts for abnormal health readings
• Inefficient management of multiple patient records and No organized system for storing medical history.

Manual health record systems also increase the burden on patients and caregivers and can lead to delayed medical decisions due to missing or incorrect health data.

Therefore, there is a need for an automated health monitoring system that can quickly record health parameters, track trends, and generate accurate health reports.

The **Personal Health Record Monitoring System** is designed to solve these problems by providing a computerized solution that automates the health data management process.

2.2 Project Overview

The **Personal Health Record Monitoring System** is a Java-based application developed to simplify the process of recording, managing, and monitoring personal health data for individuals and healthcare providers

The system allows users to:

• Enter daily health readings such as blood pressure, glucose, and heart rate
• Store and retrieve complete medical history
• Automatically detect and alert abnormal health values
• Generate organized health summary reports

2.3 Scopes and Limitations

Scope

The scope of the **Personal Health Record Monitoring System** includes the following:

• Recording and storing personal health parameters of patients
• Tracking daily health readings such as blood pressure, glucose, and heart rate
• Automatically analyzing health data and detecting abnormal values
• Generating a detailed health summary report for the patient
• Demonstrating the practical use of Java programming concepts in healthcare

The system can be used in:

• Individual patients monitoring their own health at home
• General physicians and family doctors
• Community health centers
• Nursing homes and elder care facilities
• School and college health departments

Limitations

*****The system does not connect to real-time medical devices or sensors

*It does not support online or cloud-based data storage

*The system is limited to basic health parameters and does not cover complex diagnostic data

*It does not provide direct communication between patients and doctors

*The system is designed for small-scale use and may not support large hospital databases

*It does not include advanced data encryption or full patient data security features

3. 3.SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before the actual implementation. In this phase, the system is divided into different modules, and the workflow of the application is defined. Proper system design helps in organizing the program efficiently and makes development easier.

The **Personal Health Record Monitoring System** is designed as a menu-driven application where users can select food items, enter quantities, and generate a final bill automatically.

The system is divided into several modules to perform specific tasks. Each module is responsible for handling a particular function of the system.

3.1 Modules

The main modules of the **Personal Health Record Monitoring System** are explained bel

1. Patient Record Display Module
This module is responsible for displaying the patient's personal and medical details such as name, age, blood group, medical history, and previous records. It helps the user (doctor/patient) to easily view stored health information.
Example Display:
Name: Arun Kumar
Age: 25
Blood Group: B+
Allergies: None
Last Checkup: 12/02/2026
The records are displayed when the system starts so that the user can quickly access the patient’s health details.

2.Health Data Entry Module
This module allows the user to enter and update patient health information such as symptoms, diagnosis, prescriptions, and test results.
Functions of this module include:
Accepting patient details input
Recording medical reports and prescriptions
Updating existing health records
Storing data securely
The system allows continuous updating of records to maintain accurate and up-to-date health information

3.Health Data Analysis Module
This module performs analysis of the patient’s health data and generates useful insights.
The system analyzes:
Patient vital signs (BP, sugar level, heart rate)

Medical history trends
Lab test results
Health improvements or risks
The analysis is performed automatically using system logic to help doctors and patients understand the current health condition.
Example Analysis:
Average Blood Pressure = Sum of readings / Number of readings
Sugar Level Status = Normal / High / Low
Health Report = Based on past and present data
This module helps in early detection of diseases and better decision-making for treatment.

4.Health Report Generation Module

Health Report Generation Module
This module generates the final health report for the patient based on stored medical data.
The report includes:
Patient name and details
Diagnosis information
Prescribed medicines
Test results
Doctor's remarks
Overall health status
Example Health Report Output:
------ HEALTH REPORT ------
Name: Arun Kumar
Age: 25
Blood Group: B+
Diagnosis: Mild Fever
Prescription: Paracetamol (500 mg)
Test Result: Normal
Doctor's Remarks: Patient is stable
Health Status: Good
This module helps in maintaining proper documentation and allows easy sharing of health reports with doctors and patients.

5. Exit Module

This module allows the user to safely exit the Personal Health Record Monitoring System after completing all operations such as viewing, updating, or generating health reports.
When the user selects the exit option, the system ensures that:
All patient data is saved properly
Any ongoing operations are completed
The application closes securely
The system then terminates the program and returns control to the system.
This module helps in preventing data loss and ensures safe closing of the application.

4. METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the **Personal Health Record Monitoring System**, a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:

• Problem Analysis
• System Design
• Program Development
• Testing and Validation
• Result and Documentation

4.1. Problem Analysis

In this phase, the problem is studied and analyzed in detail. The requirements of the system are identified based on the needs of individuals and healthcare providers in managing personal health records.

The main problems identified in manual billing systems are:

*Manual recording may lead to data entry errors
* Maintaining health records manually takes more time
*Difficult to track and compare health parameters over multiple days
* No automatic alerts for abnormal or critical health values

*Retrieving old health records from paper files is inefficient

*Lack of organized health summary reports for medical consultations

4.2. System Design

In this phase, the overall structure and design of the Personal Health Record Monitoring System is planned. The system is designed to include the following modules:

• Patient Profile Module – Stores patient name, age, and basic medical details
• Health Parameter Input Module – Accepts daily readings of blood pressure, glucose, heart rate, and temperature
• Data Analysis Module – Analyzes recorded values and compares them with normal ranges
• Alert Module – Generates warnings for abnormal or critical health readings
• Report Generation Module – Produces a structured health summary report for the patient

4.3. Program Development

In this phase, the actual coding and implementation of the **Personal Health Record Monitoring System** was carried out based on the system design. The program was developed using Core Java programming language to ensure reliability and efficiency in health data management.

The following steps were followed during development:

• Creation of classes and methods for health data management
• Implementation of menu-driven logic for user interaction
• Writing code for health parameter recording and analysis
• Handling user input for daily health readings
• Displaying the final health summary report and alerts

Core Java concepts such as classes, objects, loops, conditional statements, and methods were used during development.

4.4. Testing and Validation

Testing ensures that the program works correctly and produces the expected results.

Different test cases were used to check the functionality of the system.

Example Test Cases:

TEST CASE	EXPECTED RESULT
Enter patient name	Name recorded correctly
Input blood pressure reading	Correct value stored and displayed
Enter blood glucose level	Accurate reading accepted
Record heart rate	Correct pulse rate saved
Set medication reminder	Reminder triggered at correct time
Enter invalid age	Past records displayed correctly
View health history	Accurate report produced

This phase ensures that the program runs without errors and produces accurate results.

4.5. Result and Documentation

After successful testing, the results of the system are documented. The system generates the final health report based on the patient's entered vital signs and health records.

The results are demonstrated through program execution and screenshots showing the input and output of the system.

Proper documentation is prepared to explain the design, implementation, and working of the system.

4.6. Algorithm

Algorithm for the **Personal Health Record Monitoring System:**

1. Start the program.
2. Display the main menu to the user.
3. Ask the user to enter personal details(name,age,gender).
4. Enter the patient’s vital signs(blood pressure,heart rate,temperature).

5. Enter the blood glucose level of the patient.
6. Store the health records in the system.
7. Check if the entered values are within the normal range.
8. If values are abnormal,display a health alert to the user.
9. Ask the user if they want to update or view previous records.
10. If yes, retrieve and display the stored health history.
11. Generate a health summary report for the patient.
12. Display the final health report for the patient
13. End the program

4.7. Flowchart

Students should draw a flowchart showing program flow.

Example flow:

Start
|
Display Main Menu
|
Enter Patient Details(name,Age,Gender)
|
Enter Vital Signs(Blood Pressure,Heart Rate,Temperature)
|
Enter Blood Glucose Level
|
Store Health Records
|
Check if values are Normal
|
Are values Abnormal?--Yes-->Display H
| No
View Health History
|
Generate Health Report

|

Display Final Report

|

End

Students can draw using:

- draw.io
- Lucidchart
- MS Visio

5. SOURCE CODE

```
import java.util.*;

class Patient {
    int id;
    String name;
    int age;
    String bloodGroup;
```

```
    String diagnosis;

    Patient(int id, String name, int age, String bloodGroup, String diagnosis) {
        this.id = id;
        this.name = name;
        this.age = age;
        this.bloodGroup = bloodGroup;
        this.diagnosis = diagnosis;
    }

    void display() {
        System.out.println("ID: " + id);
        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
        System.out.println("Blood Group: " + bloodGroup);
        System.out.println("Diagnosis: " + diagnosis);
        System.out.println("----------------------------");
    }
}

public class PersonalHealthRecordSystem {
    static ArrayList<Patient> patients = new ArrayList<>();
    static Scanner sc = new Scanner(System.in);

    public static void main(String[] args) {
        int choice;

        do {
            System.out.println("\n--- Personal Health Record System ---");
            System.out.println("1. Add Patient Record");
            System.out.println("2. View All Records");
            System.out.println("3. Search Patient by ID");
            System.out.println("4. Exit");
            System.out.print("Enter your choice: ");
            choice = sc.nextInt();

            switch (choice) {
                case 1:
                    addPatient();
                    break;
                case 2:
                    viewPatients();
                    break;
```

```
            case 3:
                searchPatient();
                break;
            case 4:
                System.out.println("Exiting system...");
                break;
            default:
                System.out.println("Invalid choice!");
        }
    } while (choice != 4);
}

static void addPatient() {
    System.out.print("Enter ID: ");
    int id = sc.nextInt();
    sc.nextLine(); // clear buffer

    System.out.print("Enter Name: ");
    String name = sc.nextLine();

    System.out.print("Enter Age: ");
    int age = sc.nextInt();
    sc.nextLine();

    System.out.print("Enter Blood Group: ");
    String bloodGroup = sc.nextLine();

    System.out.print("Enter Diagnosis: ");
    String diagnosis = sc.nextLine();

    patients.add(new Patient(id, name, age, bloodGroup, diagnosis));
    System.out.println("Patient record added successfully!");
}

static void viewPatients() {
    if (patients.isEmpty()) {
        System.out.println("No records found.");
        return;
    }

    for (Patient p : patients) {
        p.display();
    }
```

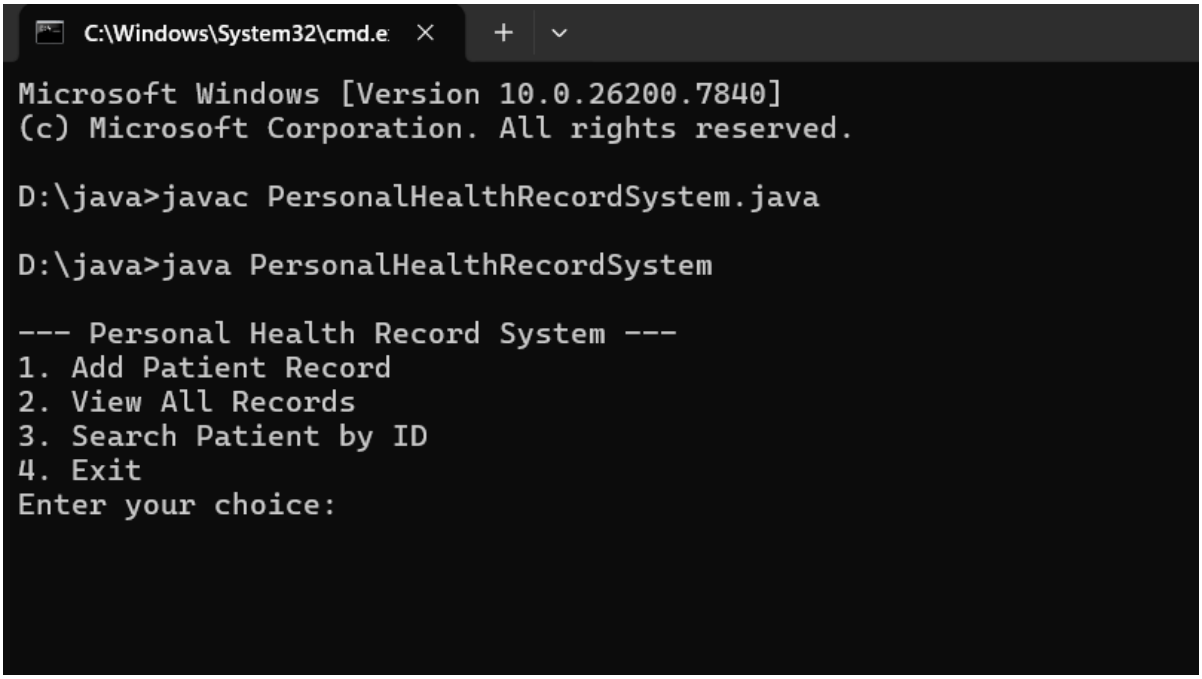
```
    }

    static void searchPatient() {
        System.out.print("Enter Patient ID to search: ");
        int id = sc.nextInt();

        boolean found = false;
        for (Patient p : patients) {
            if (p.id == id) {
                p.display();
                found = true;
                break;
            }
        }

        if (!found) {
            System.out.println("Patient not found!");
        }
    }
}
```

6. INPUT AND OUTPUT SCREENS



7. DISCUSSION & CONCLUSION

7.1 Discussion

The Personal Health Record Monitoring System was developed to automate the process of recording, managing, and monitoring personal health data for individuals and healthcare providers. The system allows users to input health parameters, track medical history, and generate health summary reports automatically

During the development process, Core Java programming concepts such as loops, conditional statements, methods, and object-oriented principles were applied. The system successfully performs health data analysis and generates accurate health monitoring reports.

The system was tested with different health parameter inputs to ensure that it performs the required calculations and analysis correctly. The results show that the system effectively detects abnormal health values, reduces manual record-keeping effort, and improves the overall efficiency of personal health monitoring

The implementation of this system proved that Java programming can be practically applied to develop simple yet effective healthcare management applications. The user-friendly interface made it easy for both patients and healthcare providers to interact with the system without requiring technical knowledge

7.2 Conclusion

The **Personal Health Record Monitoring System** successfully demonstrates the practical application of Core Java programming concepts in developing a real-world healthcare application.

The system automates the health monitoring process, reduces data recording errors, and provides faster access to patient health information. It simplifies personal health management by generating accurate health reports based on recorded parameters and alerting users about critical health conditions

This project helps students understand how Java programming can be used to develop practical software systems in the field of healthcare. The project can be further enhanced by adding features such as database connectivity, graphical user interfaces, cloud-based storage, integration with real-time medical devices, and online doctor consultation systems.

8. REFERENCES

The following resources were used during the development of this project:

Books

• Herbert Schildt – *Java: The Complete Reference*

Websites

• https://www.healthit.gov/topic/health-it-and-health-information-exchange-basics/personal health records

• https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6743674/

• https://www.who.int/health topics/digital health